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ARTIFICIAL INTELLIGENCE (SECJ3553)

SECTION 03

GROUP 2

ASSIGNMENT 3:

EcoRoute

THEME:

SMART CITY

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1.0 PEAS Model

Performance measure	1. Efficiency: Minimize total fuel consumption (L) and distance travelled (km). 2. Speed: Reduced time for collection cycles. 3. Reliability: Zero overflow incidents (preventing the "High Risk" state).
Environment	1. Roads: Urban traffic network. 2. Zones: Residential & Commercial areas. 3. Temporal: Weekdays vs. Weekends. 4. Online Platform: Cloud database & Driver App.
Actuators/Effectors	1. Display: Driver navigation interface. 2. Virtual: Route Optimization Engine (TSP). 3. Alerts: Push notifications to drivers. 4. Database: Status updates such as Set High Priority.
Sensors	1. Ultrasonic: Bin fill-level sensors. 2. GPS: Truck and Bin location trackers. 3. Weight: Truck load sensors.

P - Performance Measure

- Efficiency: The primary measure is minimizing the Path Cost. The agent is successful if it reduces the litre of diesel used per kilogram of waste collected.
- Speed: The agent aims to minimize the total duration of a collection round, ensuring drivers complete their shifts without unnecessary overtime.
- Accuracy: The agent judges its own accuracy by its ability to predict and collect waste before a bin overflows. If a bin reaches 100% capacity before a truck arrives, the agent has failed.

E - Environment

- **Roads:** The agent must navigate the actual road network of the city, which includes traffic lights, congestion, and varying distances between bins.
- **Zones:** The environment is divided into specific zones such as residential or commercial zones which dictate how fast waste accumulates.
- **Temporal:** The environment changes based on time. A weekend or public holiday environment has more waste generation than weekday.
- **Online Platform:** The agent also presents on the cloud server where it processes data from the peer-to-peer network of bins and trucks.

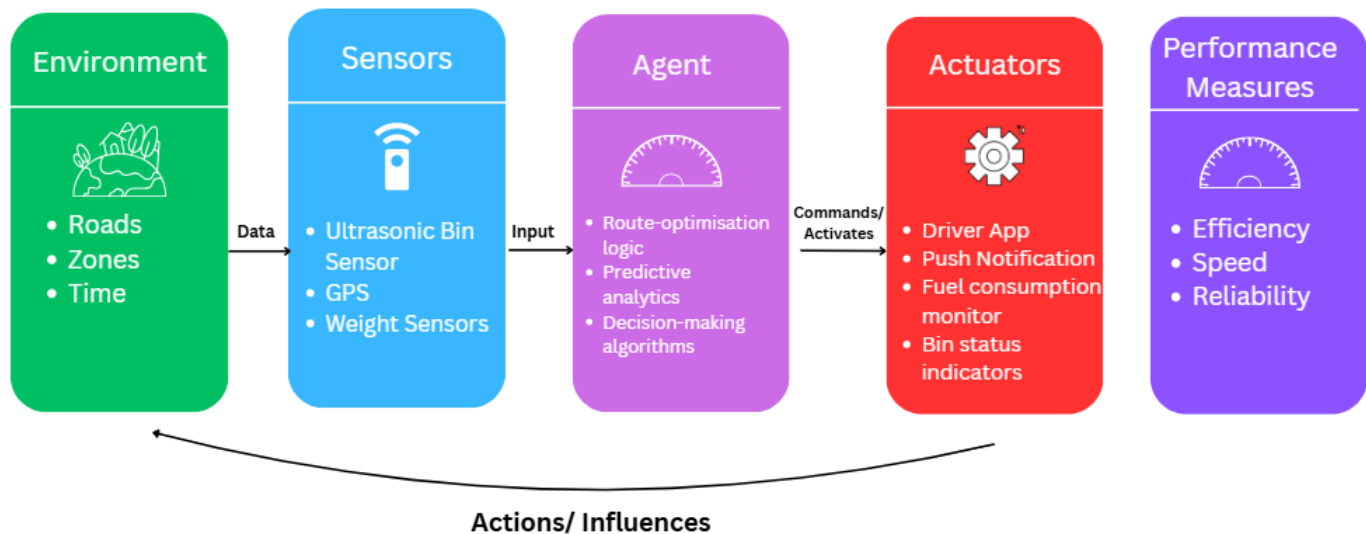
A - Actuators

- **Display:** This agent represents driver interface which displays the optimized path and turn-by-turn navigation instructions.
- **Virtual Mechanisms:** The TSP Algorithm acts as an internal actuator. It physically reorders the sequence of stops in the database for more efficient route.
- **Alerts:** The agent triggers Push Notifications when high priority bin detected. This mechanism notifies the driver to collect high priority bins.
- **Database:** The agent acts by changing the state of a bin in the system from "Monitor" to "High Priority".

S - Sensors

- **Ultrasonic Sensors:** These are the primary "eyes" inside the bin lids. They measure the distance to the waste pile to determine the Bin Fill Level.
- **GPS:** The agent uses GPS modules to perceive the coordinates of every truck and bin. This allows it to calculate proximity for the "Assign Nearest Truck" rule.
- **Weight/Load Sensors:** Sensors on the truck's suspension perceive the current weight of the vehicle, allowing the agent to detect if the truck is full.

2.0 Relations Between Each Property



The EcoRoute agent operates through a continuous cycle that integrates the PEAS parameters to achieve its optimisation objectives. The process begins in the Environment, which provides raw data perceived by sensors, such as Ultrasonic Bin Sensors, GPS, and Weight Sensors. These sensors provide real-time input to the Agent, which serves as the system's brain. Within the Agent block, the system applies Route-optimization logic, predictive analytics, and decision-making algorithms to process the incoming data.

Once a decision is reached, the Agent sends commands to activate the Actuators, which include the Driver App, Push Notifications, Fuel Consumption Monitors, and Bin Status Indicators. These actuators execute the necessary actions that directly influence the Environment. For example, by guiding a truck to a high-priority zone for collection. Finally, the entire operation is evaluated against Performance Measures, such as Efficiency, Speed, and Reliability, to ensure the results meet the system's goals. This closed feedback loop allows the agent to constantly refine its logic and decision rules, ensuring continuous improvement in smart waste management.

3.0 Representation of PEAS Properties in the Proof of Concept

In the Proof of Concept (POC) for EcoRoute, the physical components of the smart waste management system are simulated through software modules and mock data. This allows the demonstration of the AI agent's intelligent behavior without requiring physical trucks or installed sensors. The following subsections detail how each element of the PEAS model is represented within the software prototype.

3.1 Performance Measure Representation

Efficiency in the POC is measured by minimizing path cost, represented through total distance travelled and estimated fuel consumption. Distance is calculated using GPS-based route tracking, while fuel usage is estimated using a fixed fuel-per-kilometre rate.

Performance is evaluated by comparing fuel consumption per kilogram of waste collected between baseline and optimized routes.

Speed is represented by the total duration of each collection cycle. The POC records start and end times of routes and incorporates simulated traffic conditions to reflect peak and off-peak periods. The agent is considered effective if it reduces total collection time while completing routes within driver shift limits.

Reliability is measured by the system's ability to prevent bin overflow. Bin fill levels are continuously monitored, and overflow incidents are recorded if bins reach full capacity before collection. The POC demonstrates reliability when high-risk bins are collected in time, resulting in zero or reduced overflow events.

3.2 Environment Representation

The urban road network is modeled as a graph with distances and time-based traffic adjustments. Residential and commercial zones are defined with different waste generation rates, influencing bin fill behavior. The variation between weekdays and weekends affects both waste accumulation and traffic patterns.

The online platform environment is hosted on a cloud platform that integrates a central database, a route optimisation service and a driver application.

3.3 Actuator Representation

The driver interface acts as the primary actuator by displaying optimised routes. A route optimization engine functions as a virtual actuator by dynamically reordering collection stops using a TPS-based approach. Push notifications are triggered when high-priority bins are detected, alerting drivers in real time. The agent also updates bin states in the database, reflecting actions such as prioritization and collection completion.

3.4 Sensor Representation

Ultrasonic sensors are simulated to provide real-time bin fill-level readings based on zone-specific waste growth. GPS sensors track truck and bin locations, enabling distance calculation and nearest-truck assignment. Truck load sensors simulate weight changes during collection, allowing the agent to monitor capacity and adjust routes accordingly.

4.0 Behaviors of Agent To Achieve The System Goals

As an intelligent agent, the EcoRoute system always follows the cycle of perception, reasoning and action to achieve the goals of reducing operating costs, improving productivity and avoiding bin overflow. With its goal-oriented and adaptable behavior, this intelligent body can respond quickly to environmental changes.

4.1 Perception Behavior

First of all, the intelligent body uses the information obtained by the sensor to perceive the surrounding environment. The GPS sensor provides the location information of the bins and garbage collection trucks, while the ultrasonic sensor installed in the bin measures the filling percentage of the garbage bin in real time. In addition, the load sensors of the garbage truck will track the remaining capacity of the vehicle. Then, the intelligent agent can update the current status record of the system according to the sensor data.

4.2 Reasoning and Decision-Making Behavior

The agent uses the perceived data to classify the status of the bin according to the predefined knowledge representation rules. For example, when the bin's fill level exceeds the preset threshold, the agent will mark it as a high-priority bin.

After that, state space search algorithm is used to analyze the possible action sequence, and selects the sequence with the lowest path cost while considering factors such as truck availability, fuel consumption, urgency and driving distance.

The decision-making process will also be affected by environmental factors. Due to the high amount of garbage generated on weekends and holidays, the agent reduces the collection threshold. Similarly, due to the faster accumulation of garbage in commercial areas, the collection priority of garbage bins is higher than the garbage bins in residential areas.

4.3 Action Execution Behavior

After making a decision, the intelligent body uses the actuator to perform a series of actions. These actions include:

- Run the route optimization engine and rearrange the collection priority
- Assign high-priority garbage bins to the nearest available trucks
- Sending notifications and route updates to drivers
- Updating bin status in the database after collection or assigning priorities

These actions will have a direct impact on the environment and push the system to achieve the expected goal state.

4.4 Goal Achievement and Feedback Loop

After the waste collection is completed, the agent update the environment by recording the collection results to the database and resetting the bin fill level. The agent then integrates these updated data into its subsequent perception loop.

The agent maintains a feedback loop by continuously tracking the results and adjusting the system status as needed, thus, enhancing the future decision-making ability. This behavior ensures the continuous efficiency and reliability of the system while preventing the overflow of the bin.